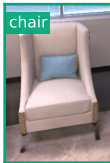


(a) an observation is a multiply-add



what



where



what $\otimes$ where



(b) the map is one vector

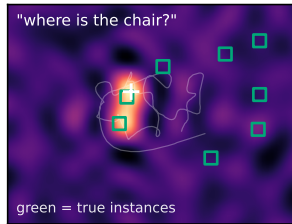


all 8,028 observations
of one walk, bundled into
8,192 phases = 32 KB

storing more
never makes it longer

(c) a question is one unbind

$\emptyset$ chair



(d) two maps add

robot A



robot B



+

A + B



= the jointly built map

to 10^{-11} , with no
data association at all

89% vs 84% for an explicit
merge and its 622 decisions